Why Is the Network Slow?

Latency, bandwidth and waiting

SCAN / LEARN

THE MAIN IDEA

Latency is the delay before data arrives. Bandwidth is how much data can travel per second. More bandwidth does not remove every delay.

FOLLOW THE SYSTEM

01 REQUEST
A tiny message leaves your device.

›

02 TRAVEL
The network adds transmission delays.

›

03 WAIT
Servers queue the request while busy.

›

04 RESPONSE
Processing and return travel take time.

LATENCY = TRAVEL + WAITING + PROCESSING

COMMON MISCONCEPTION

A fast internet plan makes every remote service instant.

MORE ACCURATE

Distance, congestion and server work still take time.

THREE QUESTIONS TO KEEP

PROBLEM: Why is one request slower than another?

FAILURE: A small delay repeated many times feels huge.

IN REAL LIFE: Gaming, video calls and remote AI.

SOLO DISCOVERY

Compare downloading a large file with waiting on a tiny answer.